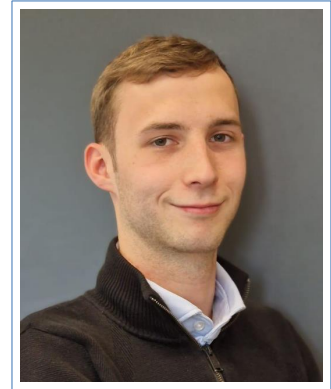


# Aaron Kreis

*M.Sc. Physics Candidate  
Expected 2027*

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*Experimental Physics · Optical Imaging · Scientific Computing*

## Research Experience

- 06 – 08/2026 **Internship – Automated Bioluminescence Microscopy and Fourier Light-Field Imaging, NMSB Group – Prof. Michael Krieg — ICFO – The Institute of Photonic Sciences**
- Expanded and characterized a multimodal microscopy platform for brightfield, fluorescence, bioluminescence, and Fourier light-field microscopy (FLFM)
  - Integrated a motorized filter wheel and designed/modified high-precision CAD models for optomechanical improvements to the microscope
  - Developed Micro-Manager plugins for automated spiral sample scanning and configurable multi-step acquisition workflows controlling camera, XY/Z stages, illumination, and filter wheel
  - Characterized the FLFM system using experimental point-spread-function measurements and reconstructed volumetric *C. elegans* data using Richardson–Lucy deconvolution; additionally evaluated deep-learning-based LUCID reconstruction models
- 07 – 10/2025 **Bachelor Thesis – Light Ray Propagation in Microscopic Imaging of Red Blood Cells, Dynamics of Fluids Group – Prof. Christian Wagner — Saarland University**
- Developed ray tracing simulations to model light propagation in microscopic imaging
  - Simulated optical interactions with rotationally symmetric biological structures
  - Performed quantitative intensity analysis on the detector plane
  - Designed a computational workflow for reproducible microscopy simulations

## Awards & Grants

- 05/2026 **Santander Scholarship for Erasmus+ Mobility** — Santander Deutschland
- 03/2026 **Travel Grant – DPG Spring Meeting (SKM)** — WE Heraeus-Stiftung
- 2025 – 2026 **Glückauf Scholarship** – RAG-Stiftung
- 03/2025 **Travel Grant – DPG Spring Meeting (SKM)** — WE Heraeus-Stiftung

## Technical Skills

Programming Python, MATLAB, C++  
Engineering CAD, 3D printing  
Tools L<sup>A</sup>T<sub>E</sub>X, Git, Linux

## Education

- 2025 – 2027 **M.Sc. Physics (Double Degree)** — Université Grenoble Alpes & Saarland University (expected)
- 2021 – 2025 **B.Sc. Physics** — Saarland University
- 2013 – 2021 **AbiBac** — Warndt-Gymnasium

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## Teaching Experience

- 2023 – 2025 **Instructor (independently taught)** — Saarland University
- Summer School “From Atoms to Zettabytes” for STEM-oriented pre-university students
  - Mathematics Preparatory Course for incoming undergraduate students
- 2022 – 2025 **Teaching Assistant for Undergraduate Physics Courses** — Saarland University
- Led problem-solving sessions accompanying undergraduate lectures
  - Guided students in experimental methods, data analysis, and scientific reporting
  - Supervised laboratory courses and tutorial sessions in mechanics, electromagnetism, and computational physics

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## Academic Service & Leadership

- 2022 – present **Student Representative** — Saarland University  
Examination Board (B.Sc./M.Sc. Physics); Appointment Committees; Faculty Council; Student Council
- 2025 – 2026 **Vice President for Partnerships** – PhITEM Student Council — Université Grenoble Alpes
- 2024 – 2025 **Founding Deputy Chair** – jDPG Regional Group Saarbrücken — German Physical Society
- 2024 **External Review Panel Member** – B.Sc. Physics Program — University of Luxembourg
- 2023 – present **Member**, German Physical Society (DPG)

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## Employment

- 2022 – 2025 **Student Assistant – Physics Outreach** — Department of Physics – Saarland University
- Designed and delivered physics demonstrations for school visits, career fairs, and public events
  - Represented the department in student recruitment initiatives
- 2021 – 2023 **Student Assistant – Process Automation** — DEKRA Automobil GmbH
- Developed automation workflows to streamline administrative processes across multiple departments
  - Reduced manual workload by optimizing repetitive tasks

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## Languages

German Native  
English  $\geq$ B2  
French  $\geq$ B2